

Would It Replicate?

Team Members: _____

Everything in this activity is simulated. The creek, its sampling sites and every chloride value come from a random number generator. It is set up like your Pennypack upstream/downstream chloride comparison, but it is **not** the Pennypack data and says nothing about your Pennypack result.

Part 1 — The Report

Team A: We measured chloride at 12 randomly chosen sites upstream of a stormwater outfall and 12 randomly chosen sites downstream. Downstream sites averaged **35 mg/L higher**. A test of the difference in means gives **p = 0.03**.

Team B will now repeat the survey exactly: same creek, same season, 12 sites on each side, with a new random choice of sites.

1. What is the chance that Team B's survey also comes out with $p < 0.05$? Each team member writes a number on their own, then compare.

Me: _____ % Teammates: _____ % _____ % _____ %

2. What did you use to get your number?

STOP. Every team member answers 1 and 2 before anyone opens the notebook.

Part 2 — Run the Creek

In the notebook you can do something no field team can: survey the same creek as many times as you like. You also know the creek's truth. Downstream sites average **33 mg/L** more chloride than upstream sites, and individual sites scatter around their average with an SD of 40 mg/L.

3. After 2.3: Run one survey three times.

Survey	Difference (mg/L)	p-value	$p < 0.05$?
1			
2			
3			

4. After 2.4 at 10, before you change it to 1000: Predict what fraction of 1000 surveys of this creek will give $p < 0.05$.

Prediction: _____ After 2.5: actual _____ smallest p _____ largest p _____

The creek never changed. What did? _____

Part 3 — Only the Surveys Like Team A's

The notebook now runs the creek in pairs: an *original* survey and a *replicate* of it. Then it keeps only the pairs whose original came out like Team A's, with p between 0.01 and 0.05.

5. **Predict, before 3.2:** Among the replicates of those kept pairs, the fraction with $p < 0.05$ will be (circle one) compared with your answer to question 4:

HIGHER ABOUT THE SAME LOWER because

6. **After 3.2:**

Pairs kept: _____ of 1000 Their replicates with $p < 0.05$: _____

All 1000 replicates with $p < 0.05$: _____

7. **After 3.3:** A survey gets $p < 0.05$ when its observed difference is more than _____ mg/L from zero. The creek's true difference is 33 mg/L. Use those two numbers to explain (a) the fraction you found in question 4, and (b) why an original $p = 0.03$ does not change the replicate's chances.

(a) _____

(b) _____

Part 4 — Twenty Surveys, Twenty Intervals

The last cell runs twenty surveys of the same creek. Each one gets a 95% bootstrap confidence interval (Class 19) and a p -value. Every team sees the same twenty surveys. Answer from the plot and table only; you do not need to read the code.

8. **Count, out of 20:**

Intervals containing 33: _____ $p < 0.05$: _____ intervals entirely above 0: _____

One survey's interval and p -value disagree about whether there is a difference. Which? _____

9. **Compare survey 3 and survey 13.** Write each one's observed difference, interval and p -value. A report that only says "significant" or "not significant" would describe them as opposites. Is that a fair description of what the two surveys found?

Survey 3: _____

Survey 13: _____

10. Back to Part 1. In one or two sentences, tell Team B what Team A's $p = 0.03$ does and does not tell them about their own survey.

If You Finish Early

A. The average observed difference among the *significant* surveys was _____ mg/L. The truth is 33. Suppose a journal only prints surveys with $p < 0.05$. What happens to the differences readers see in print?

B. With a true difference of 60 mg/L: pairs kept _____, their replicates with $p < 0.05$ _____. Why didn't the null distribution need to be rebuilt? What does this tell you about whether "the chance a $p = 0.03$ result replicates" has one answer?